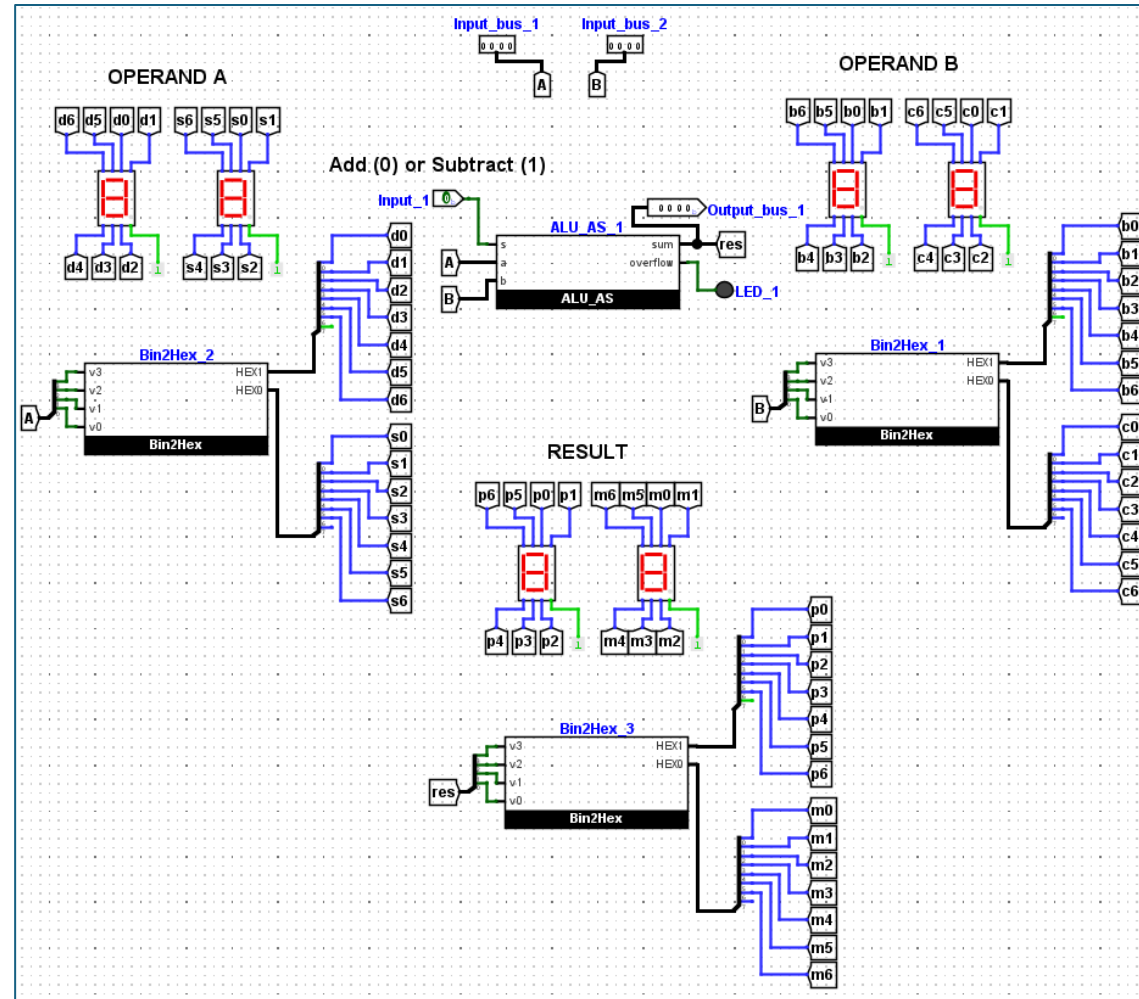


Digital Design: Lab 4

Christian Mendez

Starting Circuit

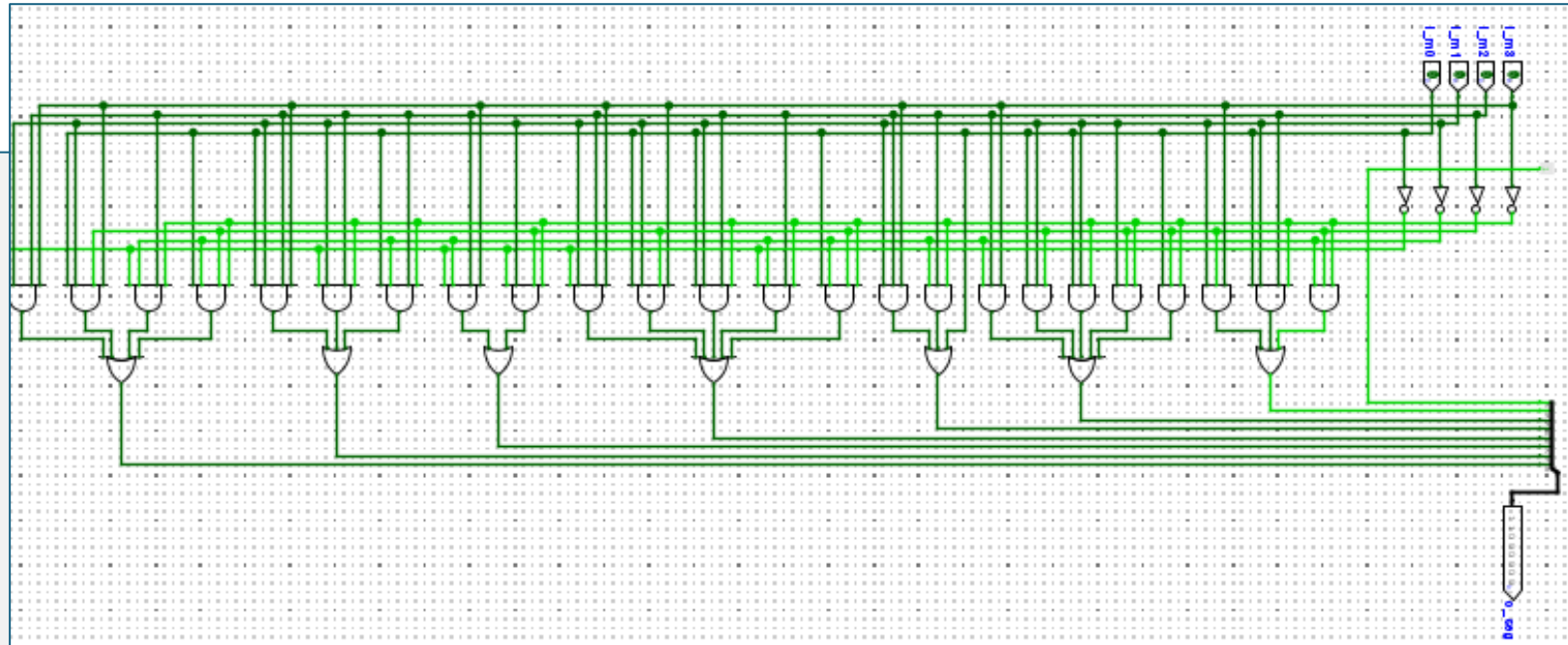
Here is the original nonfunctioning circuit we were given for the lab.



Seven Segment Display

To start fixing the circuit, we input ones and zeros according to what segments we wanted to turn on (0's are for on and 1's are off). This resulted in the following circuit.

i_m3	i_m2	i_m1	i_m0	o_seg[7..0]
0	0	0	0	1 1 0 0 0 0 0 0
0	0	0	1	1 1 1 1 1 0 0 1
0	0	1	0	1 0 1 0 0 1 0 0
0	0	1	1	1 0 1 1 0 0 0 0
0	1	0	0	1 0 0 1 1 0 0 1
0	1	0	1	1 0 0 1 0 0 1 0
0	1	1	0	1 0 0 0 0 0 1 0
0	1	1	1	1 1 1 1 1 0 0 0

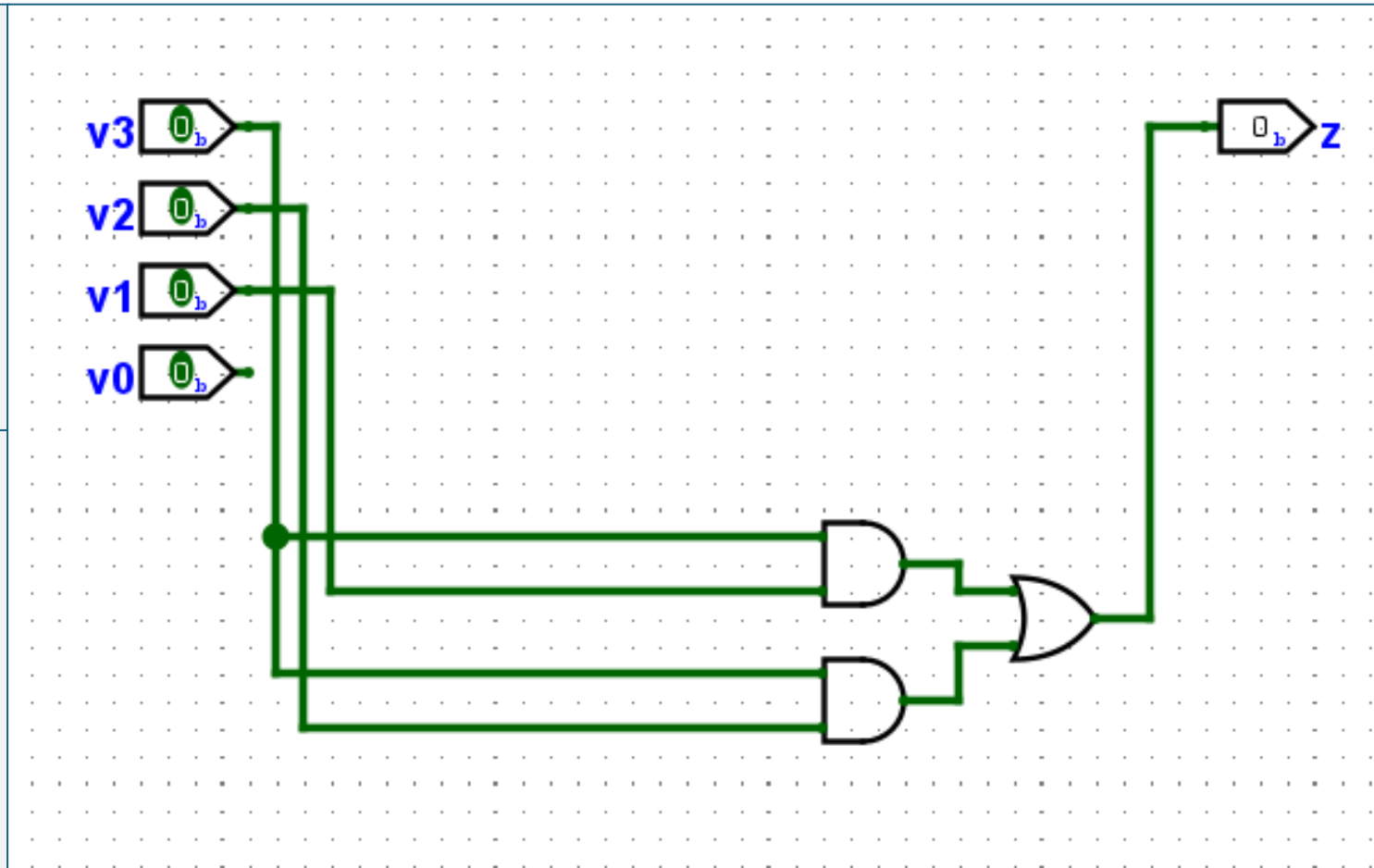


Comparator

Next, we made a circuit for our comparator. Since we need to display numbers greater than 9, we created a truth table to decide when to switch from the ones place to the tens place on the displays

v3	v2	v1	v0	z
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0
0	1	0	1	0
0	1	1	0	0

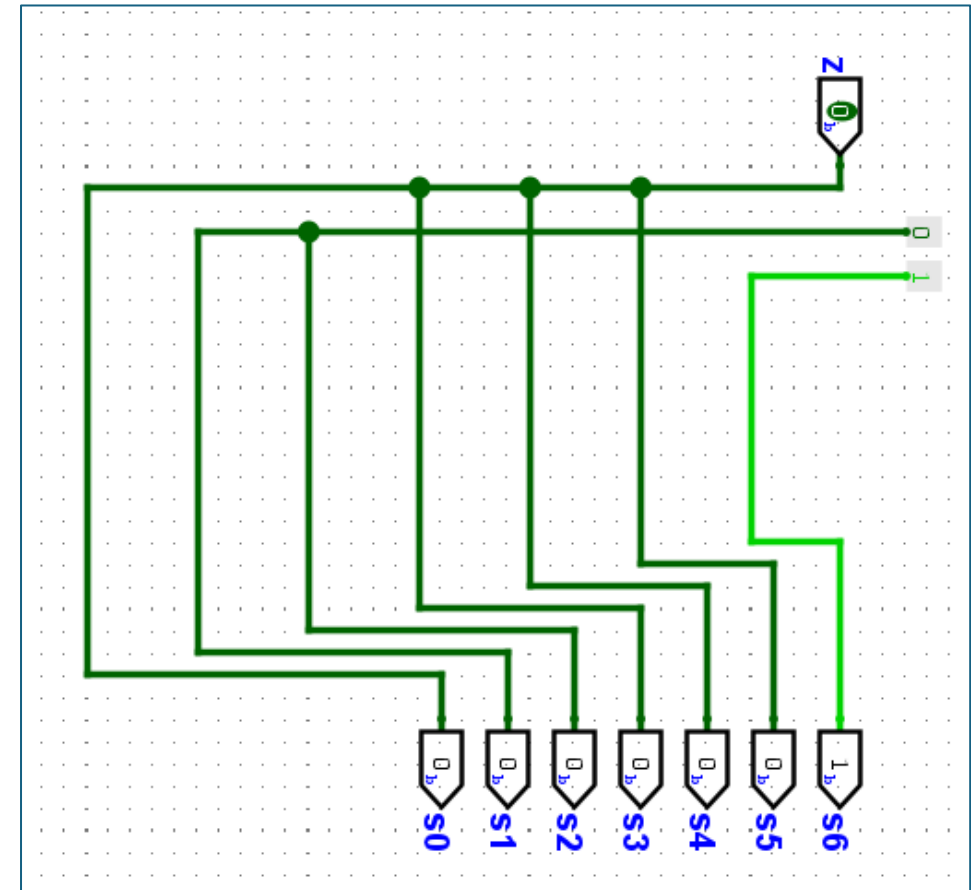
v3	v2	v1	v0	z
1	0	0	0	0
1	0	0	1	0
1	0	1	0	1
1	0	1	1	1
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	1



Circuit B

Finally, we worked on circuit B, which looks at the results from the comparator to determine whether to display a 1 or 0 in the tens place.

z	s6	s5	s4	s3	s2	s1	s0
0	1	0	0	0	0	0	0
1	1	1	1	1	0	0	1



Final Circuit

This is my final working circuit. I tested a variety of combinations of numbers, both adding and subtracting.

